Name: ______________________________ ________ / 25

Please circle your instructor's name: Kevin Meek James Gossell Margaret Short

There are 25 points possible on this quiz. No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. **[15 points]** Find the derivative of each function. You do not need to simplify your answer.

   a. $f(\theta) = \arcsin(5\theta)$

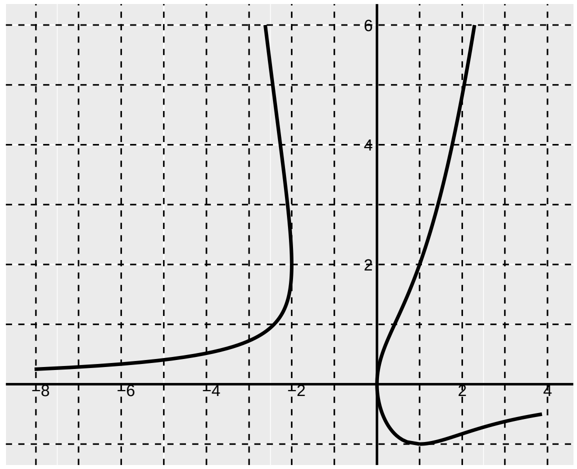
   b. $g(x) = \ln\left(\frac{x^5}{\sin(x)}\right)$

   c. $h(t) = \arctan(3t + t^2)$

   d. $f(x) = (\cos(x))^{-1} + \cos^{-1}(x)$

   e. $g(x) = e^{\ln(x)} + 5e^{3x+1} - 2^x + 6e^{21}$

2. [10 points] The graph of the curve $yx^2 = y^2 - 2x$ appears below.



a. Use implicit differentiation to find $\dfrac{dy}{dx}$.

b. Find the equation of the tangent line to the curve at $P(1,2)$.

c. Sketch the tangent line to at curve at $P(1,2)$ on the graph above.

3. **[2 points]** BONUS: Let $f(x) = x^{3x}$, $x > 0$. Find $f'(x)$.